

Hyperparameters in Deep Neural Networks

CS4152 — Deep Learning and Neural Networks

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Outline

- 1 Parameters vs. Hyperparameters
- 2 Additional Hyperparameters
- 3 Why Tuning Is Empirical
- 4 Domain and Time Dependence
- 5 Conclusion

Model Parameters

- Learned during training:
 - Weights $W^{[l]}$
 - Biases $b^{[l]}$
- Updated via gradient descent.

Hyperparameters

- Set before training; govern the learning process.
- Examples:
 - Learning rate α
 - Number of iterations
 - Number of layers L
 - Number of hidden units $n^{[l]}$
 - Activation functions
- Control how W and b evolve.

More Advanced Hyperparameters

- Introduced in later courses:
 - Momentum
 - Mini-batch size
 - Regularization coefficients
 - Adam/RMSProp parameters
- Deep learning has many hyperparameters compared to classic ML.

Empirical Nature of Hyperparameter Tuning

- Hard to predict best values in advance.
- Typical workflow:
 - Pick a value (e.g., $\alpha = 0.01$)
 - Train and observe cost J
 - Adjust based on observed behavior
- Many trials required to find good combinations.

Learning Rate Examples

- Small α : slow convergence.
- Large α : divergence.
- Good α : fast and stable decrease of J .

Cross-Domain Differences

- Optimal hyperparameters differ across:
 - Computer vision
 - Speech recognition
 - NLP
 - Structured data tasks
- Intuition does not always transfer.

Hyperparameters Change Over Time

- Improvements in:
 - Hardware (GPUs/CPU)
 - Data availability
 - Software libraries
- May require re-tuning every few months.
- Best practice: periodically revisit hyperparameter choices.

Key Takeaways

- Hyperparameters shape training behavior and outcomes.
- Deep learning involves many hyperparameters—not just α .
- Tuning is inherently empirical; expect iteration.
- Re-tune as data, hardware, and objectives evolve.